

Polynomial lower bounds for every fixed Kreiss constant

Candidate full solution to Question 6.1 of arXiv:2512.10025

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Status. Candidate full solution, likely valid; novelty unconfirmed. The argument answers Question 6.1 affirmatively. It does not answer the stronger uniform exponent Question 6.2.

1 Source question

The source is N. Chalmoukis, G. Tsikalas, and D. Yakubovich, *Operators with small Kreiss constants*, arXiv:2512.10025v2 [1]. For an $N \times N$ matrix A , write

$$\mathcal{K}(A) = \sup_{|z|>1} (|z| - 1) \|(zI - A)^{-1}\|.$$

The source defines $P(N, K)$ as the least universal bound on $\sup_{n \geq 1} \|A^n\|$ over all $N \times N$ matrices with $\mathcal{K}(A) \leq K$.

Question 6.1 on PDF page 18 asks whether there is a strictly increasing $\alpha : (1, \infty) \rightarrow (0, 1)$, with endpoint limits 0 and 1, such that

$$P(N, K) \gtrsim_K N^{\alpha(K)} \quad (N \geq 1, K > 1).$$

$$\leq \frac{1}{|\lambda| - 1} + C \frac{|\log(|\lambda| - 1)|^{-1/2}}{|\lambda| - 1},$$

where the last inequality is due to (5.4), with a replaced by $1/|\lambda|$. In other words, F satisfies (1.9) with $\varepsilon(x) = C(\log(1/x))^{-1/2}$.

6. OPEN QUESTIONS

Nikolski conjectured [23] that one should always be able to obtain sublinear growth in N in (1.2) as long as the Kreiss constant remains bounded. In fact, he proved that if A is a diagonalizable $N \times N$ matrix with unimodular eigenvalues, then there exists $\varepsilon > 0$ (depending on the basis constant of the eigenvectors of A) such that

$$\sup_{n \geq 1} \|A^n\| \leq 2\pi K N^{1-\varepsilon}.$$

Unfortunately, it is not clear how exactly ε depends on K here (see also [23] Theorem 2.9.9), while the case of general $N \times N$ matrices currently lies out of reach of Nikolski's methods. As such, there remains a sizeable gap between the lower bounds (1.5)-(1.6) and Spijker's estimate (1.2), which naturally motivates the following questions:

Question 6.1. Does there exist a strictly increasing function $\alpha(\cdot) : (1, \infty) \rightarrow (0, 1)$ with $\lim_{K \rightarrow 1+} \alpha(K) = 0$ and $\lim_{K \rightarrow \infty} \alpha(K) = 1$ such that

$$P(N, K) \gtrsim_K N^{\alpha(K)},$$

for all $N \geq 1$ and $K > 1$?

Figure 1: The source context and complete statement of Question 6.1, PDF page 18.

2 Proof intuition

The source obtains arbitrarily high powers of $\log N$ by increasing an internal matrix dimension, but its perturbation constants deteriorate too quickly with that dimension. The key idea here is to tensorize a single 2×2 nilpotent perturbation instead. Tensor products amplify a carefully selected power exponentially in the tensor degree m ,

while a pointwise comparison collapses the entire perturbation estimate to the scalar weight $((j+n)/j)^a - 1$. Its averaged square is $O(a^2)$ independently of m . Abel summation then gives a Kreiss constant $1 + O(a)$, whereas choosing $m \asymp a \log L$ in an L -coordinate compression turns the exponential tensor amplification into a positive power of the total dimension.

3 Quantitative theorem

All logarithms are natural. For $0 < a < \frac{1}{2}$, set

$$\gamma(a) = a \sqrt{\frac{6}{(1-2a)(1-a)}}, \quad \beta(a) = \frac{a \log 2}{4(1+a \log 2)}.$$

Theorem 1. *For every $0 < a < \frac{1}{2}$ there is $c_a > 0$ such that*

$$P(N, 1 + \gamma(a)) \geq c_a N^{\beta(a)} \quad (N \geq 1).$$

Consequently, Question 6.1 has an affirmative answer.

The proof is self-contained except for the previously known large- K lower bound of Nikolski [2], already recorded as equation (1.5) in the source paper [1].

4 A dimension-free perturbation estimate

Let

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J^2 = 0.$$

Fix $m \geq 1$ and put $H_m = (\mathbb{C}^2)^{\otimes m}$. On $\ell^2(\mathbb{N}; H_m)$ define the weighted backward shift $T = T_{a,m}$ by

$$(Tx)_j = B_{m,j+1} x_{j+1}, \quad B_{m,k} = \left(I_2 + \frac{a}{m} \log \frac{k}{k-1} J \right)^{\otimes m} \quad (k \geq 2).$$

Let S be the unweighted backward shift on the same space.

Because tensor products multiply componentwise and $J^2 = 0$, the block of T^n carrying coordinate $j+n$ to coordinate j is exactly

$$C_{j,n} = \left(I_2 + \frac{a}{m} \log \frac{j+n}{j} J \right)^{\otimes m}. \quad (1)$$

For $t \geq 0$, telescoping the difference of tensor powers gives

$$\|(I_2 + tJ)^{\otimes m} - I_{H_m}\| \leq (1+t)^m - 1 \leq e^{mt} - 1.$$

Hence (1) implies

$$\|C_{j,n} - I_{H_m}\| \leq \left(\frac{j+n}{j} \right)^a - 1. \quad (2)$$

Lemma 1. *With $D_n = T^n - S^n$, one has, for every $x \in \ell^2(\mathbb{N}; H_m)$ and $M \geq 1$,*

$$\frac{1}{M} \sum_{n=1}^M \|D_n x\| \leq \gamma(a) \|x\|.$$

The bound is independent of m .

Proof. Put $f_a(u) = (u^{-a} - 1)^2$. A direct integration gives

$$I_a := \int_0^1 f_a(u) du = \frac{2a^2}{(1-2a)(1-a)}. \quad (3)$$

By (2), changing variables from (j, n) to $k = j + n$ yields

$$\frac{1}{M} \sum_{n=1}^M \|D_n x\|^2 \leq \sum_{k \geq 2} q_{M,k} \|x_k\|^2,$$

where

$$q_{M,k} = \frac{1}{M} \sum_{n=1}^{\min(M, k-1)} \left[\left(\frac{k}{k-n} \right)^a - 1 \right]^2.$$

We claim $q_{M,k} \leq 3I_a$. If $k \leq 2M + 1$, monotonicity of f_a and the upper Riemann-sum estimate give

$$q_{M,k} \leq \frac{1}{M} \sum_{j=1}^{k-1} f_a(j/k) \leq \frac{k}{M} I_a \leq 3I_a.$$

If $k > 2M + 1$, then $1 - n/k > 1/2$ for $1 \leq n \leq M$. The mean value theorem gives

$$(1 - n/k)^{-a} - 1 \leq a 2^{a+1} \frac{n}{k},$$

and therefore

$$q_{M,k} \leq a^2 2^{2a+2} \left(\frac{M}{k} \right)^2 \leq a^2 2^{2a} < 2a^2 \leq 3I_a.$$

Thus the quadratic average is at most $3I_a \|x\|^2$. Cauchy–Schwarz in n and (3) finish the proof because $\sqrt{3I_a} = \gamma(a)$. $\square$

Proposition 1. $\mathcal{K}(T_{a,m}) \leq 1 + \gamma(a)$ for every $m \geq 1$.

Proof. The block formula gives $\|T^n\| \leq (n+1)^a$, so the Neumann resolvent series converges in norm for $|z| > 1$. Let $r = |z|$, $q = r^{-1}$, and $d_n = \|D_n x\|$. Lemma 1 says that $A_N := \sum_{n=1}^N d_n \leq \gamma(a)N\|x\|$. Abel summation gives

$$\begin{aligned} (r-1) \sum_{n \geq 1} r^{-n-1} d_n &= (1-q) \sum_{n \geq 1} q^n d_n \\ &= (1-q)^2 \sum_{N \geq 1} q^N A_N \leq \gamma(a)q\|x\| \leq \gamma(a)\|x\|. \end{aligned}$$

Since S is a contraction,

$$(r-1) \|(zI - S)^{-1}\| \leq (r-1) \sum_{n \geq 0} r^{-n-1} = 1.$$

Writing the resolvent of T as the resolvent series for S plus the series of the D_n proves the assertion. $\square$

5 Finite compressions and polynomial growth

Let $A_{m,L}$ be the restriction of $T_{a,m}$ to its invariant first L coordinates. It acts on a space of dimension $2^m L$, is nilpotent, and by Proposition 1 satisfies

$$\mathcal{K}(A_{m,L}) \leq 1 + \gamma(a).$$

Apply $A_{m,L}^{L-1}$ to the unit vector supported at coordinate L with internal component $e_2^{\otimes m}$. Formula (1) gives

$$\|A_{m,L}^{L-1}\| \geq \left(1 + \left(\frac{a \log L}{m} \right)^2 \right)^{m/2}. \quad (4)$$

If $a \log L \geq 2$ and $m = \lfloor a \log L \rfloor$, then $m \geq a \log L/2$ and $a \log L/m \geq 1$. Therefore

$$\|A_{m,L}^{L-1}\| \geq 2^{m/2} \geq L^{a \log 2/4}, \quad \dim A_{m,L} = 2^m L \leq L^{1+a \log 2}. \quad (5)$$

Given a sufficiently large target dimension N , take

$$L = \left\lfloor N^{1/(1+a \log 2)} \right\rfloor, \quad m = \lfloor a \log L \rfloor.$$

Then $2^m L \leq N$. Pad $A_{m,L}$ with a zero block to obtain an $N \times N$ matrix without increasing its Kreiss constant. The floor losses are absorbed in a constant depending only on a , and (5) yields

$$P(N, 1 + \gamma(a)) \gtrsim_a N^{\frac{a \log 2}{4(1+a \log 2)}}.$$

The finitely many smaller N are absorbed using $P(N, K) \geq 1$. This proves the quantitative assertion of Theorem 1.

6 The exponent requested in Question 6.1

For $K > 1$, put

$$a(K) = \frac{K-1}{4K} \in (0, 1/4).$$

On $0 < a \leq 1/4$, one has $\gamma(a) \leq 4a$, and hence $1 + \gamma(a(K)) < K$. Since $P(N, K)$ increases with K , the theorem gives a positive polynomial exponent at every $K > 1$.

Nikolski's established estimate [2], quoted in the source as equation (1.5), is

$$P(N, K) \geq \frac{1}{5} N^{1-2/K} \quad (K > \pi + 1).$$

Define

$$\alpha(K) = \begin{cases} \frac{a(K) \log 2}{4(1 + a(K) \log 2)}, & 1 < K \leq \pi + 1, \\ 1 - \frac{2}{K}, & K > \pi + 1. \end{cases}$$

The first branch is strictly increasing and tends to 0 at 1^+ . Its value is less than $\log 2/16$, whereas the right limit of the second branch at $\pi + 1$ is $1 - 2/(\pi + 1) > 1/2$. Thus the piecewise function is strictly increasing on $(1, \infty)$, and its limit at infinity is 1. The two lower bounds prove $P(N, K) \gtrsim_K N^{\alpha(K)}$, completing the affirmative answer.

7 Verification and limitations

The script `code/verify_tensor_shift.py` checks the tensor product identity, the scalar averaging inequality on finite grids, the power witness, and sampled Kreiss ratios for three finite matrices. All tests passed. The sampling is not a proof of the resolvent estimate; Proposition 1 is the proof.

The four lightweight run indexes and bounded arXiv/web searches were queried using arXiv:2512.10025 and combinations of “small Kreiss constants”, “ $P(N, K)$ ”, “polynomial lower bound”, “tensor power”, and “weighted shift”. They returned the source and background material but no later answer to Question 6.1. The current arXiv version is v2 (11 March 2026), where the question remains open. This is not an exhaustive literature review; novelty is unconfirmed.

The most important human-review points are Lemma 1, the Abel conversion in Proposition 1, and the exact-dimension padding. Question 6.2 remains open.

References

- [1] N. Chalmoukis, G. Tsikalas, and D. Yakubovich, *Operators with small Kreiss constants*, arXiv:2512.10025v2, 2026.
- [2] N. Nikolski, *Sublinear dimension growth in the Kreiss matrix theorem*, Algebra i Analiz 25 (2013), no. 3, 3–51; St. Petersburg Math. J. 25 (2014), 361–396.